

Prediction model development for petroleum refinery wastewater treatment



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ABSTRACT

Multi-stage biological treatment of petroleum refinery wastewater using different biological conditions (anaerobic–anoxic–aerobic) has many advantages over other biological methods. It can result in maximum treatment for type of complex wastewater. In this study, raw data obtained from two multi-stage biological reactors (MSBR) used for treatment of different loads of petroleum refinery wastewater was used for developing mathematical model that could predict the process trend. The data consists of 160 entries and were gathered over approximately 180 days from two MSBR reactors that were continuously operated in parallel. A Matlab code was written with two configurations of artificial neural network. The configurations were compared and different number of neurons at the hidden layer were tested for optimum model that represent the process behavior under different loads. The tangent sigmoid transfer function (Tansig) at hidden layer and a linear transfer function (Purelin) at output layer with 6 neurons were selected as the optimum best model. The model was then used for prediction; highest removal efficiency observed was 98% which was repeatedly recorded for various loads. Effluent concentration below 100 mg/L as chemical oxygen demand (COD) was recorded for influent concentration ranged between 900 and 3600 mg COD/L.

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1. Introduction

1.1. Petroleum refinery wastewater treatment

The global demand for petroleum products increases rapidly, as a result our environment is exposed to rising hazardous impact. One of these problems is created by the huge amount of wastewater generated during refining of crude oil, where approximately 3.5–5 m³ of wastewater per ton of crude oil generated contains high levels of pollutants [1]. These organic and inorganic pollutants, such as ammonia, oil, phenol, sulphur based compounds and heavy metals, poses hazardous to the environment [2]. Despite of the several existing conventional treatments processes of petroleum refinery wastewater, they often do not meet today's stringent standards because of the high concentrations of contaminants, low efficiency or high costs of treatment [3]. Several parameters could be used to monitor wastewater generated in petroleum refineries. One of them is the chemical oxygen demand (COD) which requires

relatively shorter time [4] and can be used to assess the overall quality.

Multi-stage biological treatment could be achieved by alternating the anaerobiosis and aerobiosis phases. This phenomenon is used mainly in nutrient removal [5]. Multi-stage biological reactor (MSBR) employing anaerobic condition followed by anoxic and aerobic, works on the basis of biological COD, nitrogen (N) and phosphorous (P) removal which is achieved by manipulating three sets of biochemical reactions; nitrification, denitrification and biological excess phosphorus removal processes [6]. Anaerobic–anoxic–aerobic process is applied widely in wastewater treatment plants. Their advantages [7,8] include; high efficiency of pollutant removal, stable performance, ability to stand shock loading, aeration energy savings, the ability to produce sludge that settles well by using nitrate to oxidize influent biochemical oxygen demand (BOD), the preanoxic process requires less oxygen for aeration, and eliminates the cost of methanol addition.

1.2. Artificial neural networks

Modeling and prediction of a process can minimize the efforts to create new data by using existing historical experimental data

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to predict unknown data [9]. One of the techniques is artificial neural networks (ANN) which is a mathematical modeling software used to simulate complex relationships. It has been used in application of engineering, pattern recognition and analysis [10] as well as in medical complex prediction [11]. Artificial neural networks is a non-linear control strategy that is used to simulate a non-linear biological treatment process when a large amount of data are available but no reliable model and little knowledge of how the process works [12]. ANN have a structure that composed of highly interconnected processing elements known as neurons, working together to process the information similar to the brain nervous system [13]. One of its advantages is that ANN do not require any prior knowledge about the structure of the relationships that exist between controlling parameters in the treatment process [14].

The ability of ANN for trend generalization reveals that ANN model is a useful tool for prediction of data [15,16]. Generally, the structure of a neural network is formed by an input layer, one or more hidden layers, and the output layer or can be summarized as (input, hidden node, output). The number of neurons in a layer and the number of layers depends strongly on the complexity of the problem studied [Amato]. The input layer receive the data and transfer them to neurons in the hidden layer where they mathematically processed and the result is transferred to the neurons in the last layer which provide the network's output. Various transfer functions are available [17–19], some of the most commonly used activation functions are three. The first is the linear function, $g(x)=x$. It is obvious that the input units use the identity function. Sometimes a constant is multiplied by the net input to form a linear function. The second is sigmoid function, $g(x)=1/(1+e^{-x})$. This function is especially advantageous for use in neural networks trained by back-propagation, because it is easy to differentiate, and thus can dramatically reduce the computation burden for training. It applies to applications whose desired output values are between 0 and 1. The third is hyperbolic tangent function $g(x)=((e^x)-(e^{-x}))/((e^x)+(e^{-x}))$. This function has similar properties to the sigmoid function. It works well for applications that yield output values in the range of -1 and 1 .

The literature is rich with many application for ANN in wastewater treatment to model the existing data and predict the performance. ANN with configuration of tan sigmoid activation function for the input and hidden layers, and linear activation function for the output activation function, in the back propagation neural network with 14 neurons and Levenberg–Marquardt back propagation training algorithm (TrainLM) predicted the actual experimental results for COD removal with correlation coefficient (R^2) of 0.997 and mean square error (MSE) of 0.000376 [20]. Modeling of laboratory data of ten months duration for full working wastewater treatment plant using ANN with tan sigmoid activation function for the input and hidden layers, while the linear activation function was used as the output activation function, resulted in R^2 values ranged from 0.63 to 0.81 for BOD, and from 0.45 to 0.65 for suspended solids (SS) [10]. These successful applications of ANN in wastewater treatment encouraged the researchers to use it for their developed treatment systems. MSBR for treatment of petroleum refinery wastewater was specifically configured to treat this type of complex wastewater with the over 90% efficiency [21].

Petroleum refinery wastewater (7300 mg COD/L) batch from a local petroleum refinery was treated in two MSBRs operated in parallel under seven organic volumetric loading rates for over 180 days. Two parallel MSBR reactors were used to evaluate the treatment efficiency of petroleum refinery wastewater with COD removal found to be in the range of 98–97% [21].

The present study follows from the previous investigation by modeling the MSBR treatment of petroleum refinery wastewater considering the influent and effluent COD concentration under

different loads. The developed model was then used to simulate the reactor to determine the optimum performance.

2. Methodology

2.1. Experimental data

The original raw data were adapted from previous work [21], in which the data represented two laboratory-scale MSBR reactors that were operated in parallel (A and B) at room temperature to treat petroleum refinery wastewater. The raw petroleum refinery wastewater was collected from a local petroleum refinery and fed to the two reactors in different concentration ranging on average from 982 to 6972 mg COD/L over approximately 180 days. COD was the key parameter and was tested for influent and effluent samples following colorimetric method using a HACH DR 2000 spectrophotometer. Other parameters were monitored to maintain proper biological condition and were measured according to standard methods [22]. The COD data gathered from this experiment were 160 entries for influent and effluent each.

2.2. Modeling procedure and data preparation

The MSBR's COD monitoring results during different loading were used for modeling of process trend. Artificial neural network in MATLAB (R2009a) was used as a mathematical software to simulate and predict the pattern of the reactor. Optimal generalization was targeted from this software and the Levenberg–Marquardt algorithms were used as training function and batch gradient descent with momentum back propagation algorithms (TrainGDM) as adaption learning function, feed-forward back propagation network type was selected. The fastest algorithm for this problem is the Levenberg–Marquardt algorithm. On the average, it is over four times faster than the next fastest algorithm. This is the type of problem for which the LM algorithm is best suited—a function approximation problem where the network has fewer than one hundred weights and the approximation must be very accurate and the LM algorithm decreases much more rapidly with time than the other algorithms [23]. The software was used to write a modeling code to compare two configurations (Logsig–Purelin and Tansig–Purelin) and to identify the best structure that represents the treatment system. The first configuration was Log sigmoid transfer function (Logsig) at hidden layer and a linear transfer function (Purelin) at output layer. And the second configuration was Tangent sigmoid transfer function (Tansig) at hidden layer and a linear transfer function (Purelin) at output layer. As the raw data was obtained from biological process that has continuous valued targets, thus, linear activation function (Purelin) was used for both configurations for the output neuron [10]. This research consider only one hidden layer, the architecture of ANN are input layer, one hidden layer and output layer.

The raw data was normalized for input and output by dividing all the input data with the maximum input and the output data with the maximum output; this resulted in the data to be in the range between 0 and 1. In this code, different number of neurons (5–35) in the hidden layer was tested to determine the number of neurons that results in a better learning performance without limiting the ability of the neural network to model the process and without losing the training trend generalization [14]. The model was run at 100 times for every neuron tested for better initialization. The minimum root mean square error (RMSE), maximum variance accounted for (VAF), maximum R^2 and minimum mean absolute percentage error (MAPE) were calculated for every run to define the optimum number of neurons.

Table 1
Number of neurons tested and the score for evaluation parameters.

Neurons	Logsig–Purelin				Tansig–Purelin			
	RMSE	VAF	R^2	MAPE	RMSE	VAF	R^2	MAPE
5	0.103	77.177	0.770	16.645	0.104	76.897	0.767	16.870
6	0.101	77.850	0.778	16.508	0.100	78.465	0.784	17.547
7	0.100	78.453	0.784	16.303	0.101	78.283	0.781	16.562
8	0.101	78.161	0.782	15.858	0.102	77.752	0.777	16.908
9	0.100	78.569	0.785	16.446	0.100	78.205	0.782	15.906
10	0.100	78.371	0.784	16.413	0.101	78.322	0.782	16.296
11	0.100	78.580	0.785	16.658	0.100	78.295	0.783	17.028
12	0.100	78.313	0.783	16.599	0.101	78.326	0.781	16.570
13	0.098	79.081	0.791	16.133	0.101	78.161	0.782	16.618
14	0.099	78.662	0.786	15.972	0.101	78.315	0.779	15.821

Italic indicates: minimum RMSE, maximum VAF, maximum R^2 and minimum MAPE.

3. Results and discussion

3.1. Modeling results

The Multi-stage Biological Reactor was modeled using artificial neural networks software. Logsig–Purelin transfer function was compared to Tansig–Purelin transfer function to define the optimum model. The selected model was then used to predict the reactor performance.

During testing and validation of data, the number of neurons in the hidden layer was tested ranging from 5 to 35. Table 1 below shows the number of neurons tested and the score registered for RMSE, VAF, R^2 and MAPE during evaluation of Logsig–Purelin, and Tansig–Purelin transfer functions.

Although the number of neurons in the hidden layer are in the range of 5–35, but from Fig. 1 it was noted that after neuron 14 and from plotted line representing the R^2 , the training set is losing similarity with R^2 from the validation set. That indicate over fitting

resulting in a model that not able to generalize the pattern of the data that was used as training set during validation [24].

Thus, the number of neurons in the hidden layer was limited to the range between 5 and 14 neurons, and the optimum neuron was selected as shown in Table 1 based on minimum RMSE, maximum VAF, maximum R^2 and minimum MAPE.

Logsig–Purelin transfer function indicated 13 neurons is the optimum, while Tansig–Purelin transfer function suggested 6 neurons. It is usually preferable to use of simpler models, with fewer number of parameters than more complicated ones with more parameters, whenever feasible [10]. Thus, tangent sigmoid transfer function (Tansig) at hidden layer and a linear transfer function (Purelin) at output layer with 6 neurons is the optimum transfer function. The RMSE, VAF and R^2 values resulted with six neurons were the indicators for the modeling accuracy of the selected architecture. Compared to similar indicators with 13 neurons in the other architecture. Figs. 2 and 3 showed the measured experimental

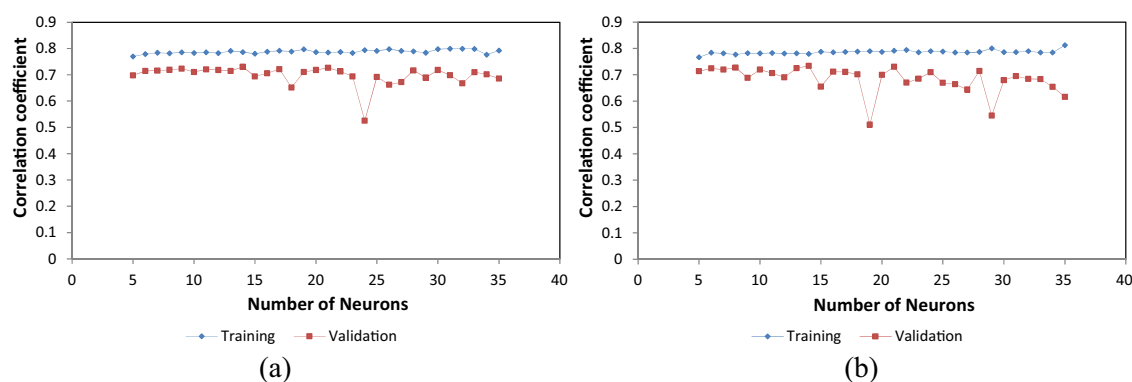


Fig. 1. R^2 scores versus number of neurons tested for (a) Logsig–Purelin transfer function, and (b) Tansig–Purelin transfer function.

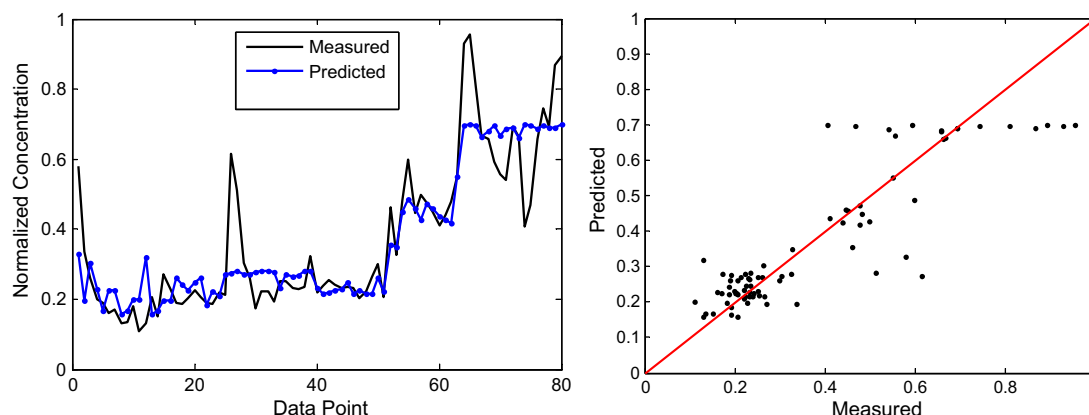


Fig. 2. MSBR measured and predicted normalized data for training set.

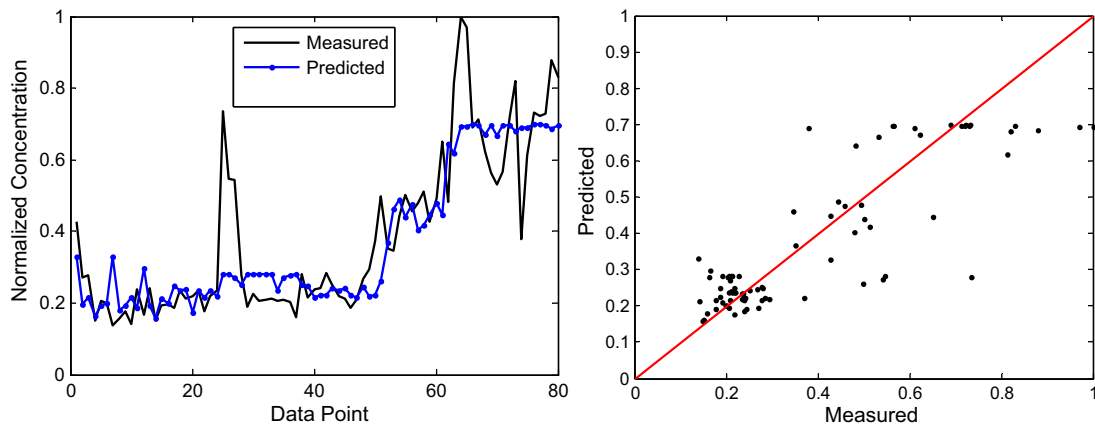


Fig. 3. MSBR measured and predicted normalized data for validation set.

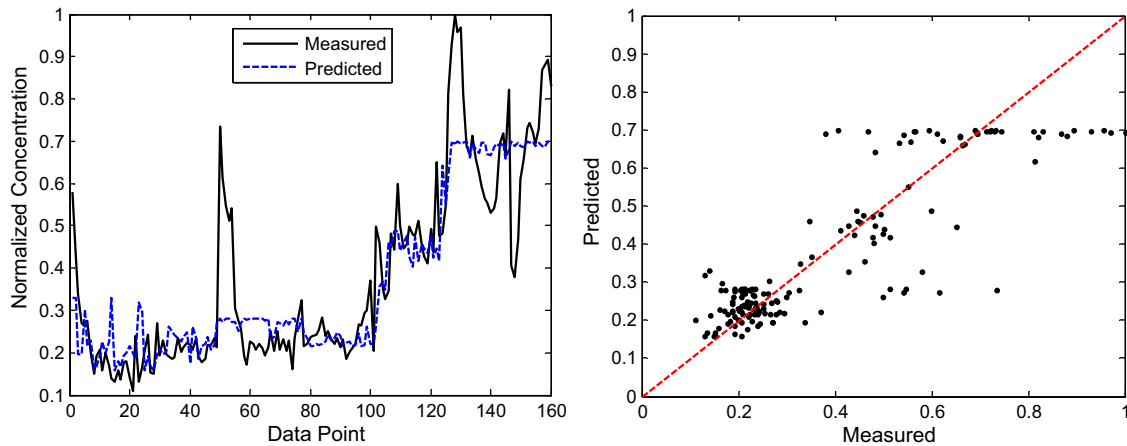


Fig. 4. MSBR measured and predicted normalized data for actual data simulation.

and predicted data using ANN for eighty entries of data that were used for training and validation respectively. The best selected model shows significant prediction of actual experiment as in Table 1 for RMSE, VAF and R^2 results; hence, it was then used for prediction.

3.2. Prediction results

The best model with Tansig–Purelin transfer function and 6 neurons was used to simulate random data to find out the optimum

efficiency. Fig. 4 shows all the hundred and sixty data set that was used for both the training and validation, used here for prediction.

Random data entries ranged from 500 to 10,000 mg/L was used as influent to simulate the reactor performance; Fig. 5 shows the predicted influent and effluent concentrations in addition to removal efficiency. Highest removal efficiency observed was 98% recorded at range of influent between 255 and 3200 mg COD/L and of influent in the ranged between 5200 and 6300 mg COD/L and of influent from 7300 mg COD/L up to the last value tested. Effluent concentration below 100 mg COD/L was recorded for influent concentration ranged between 900 and 3600 mg COD/L.

4. Conclusion

Raw data from petroleum refinery wastewater treatment with different loads using two MSBR reactors were successfully used for modeling. Modeling the data obtained from the multi-stage biological reactor under various loads, resulted in a model that used tangent sigmoid transfer function (Tansig) at hidden layer and a linear transfer function (Purelin) at output layer with 6 neurons as the optimum transfer function.

Prediction using the optimum model with random data entries ranged between 500 and 10,000 mg/L as influent, resulted in a pattern that simulate the reactor performance for new data other than the data that experimentally tested in the lab. Lab experiment was showing highest removal of 97% and could not define the exact load that can give effluent concentration below 100 mg/L; but from the prediction model, 98% removal efficiency was repeatedly

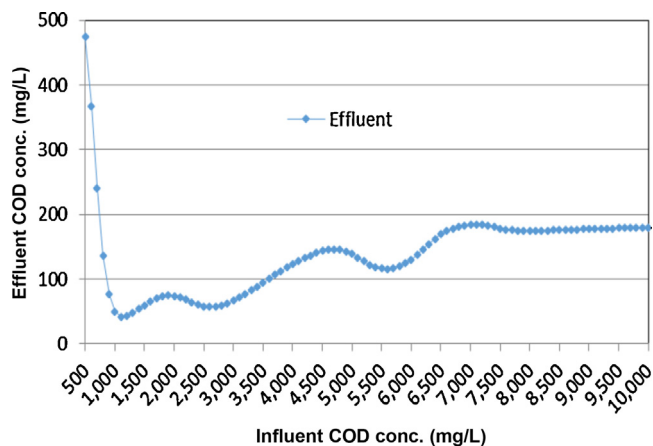


Fig. 5. Influent versus effluent data using best selected model with Tansig–Purelin transfer function.

recorded for various loads, and loads that can give concentration below 100 mg COD/L were defined.

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